Sreyoshi Basu

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Education

2022–2024 **M.A. Linguistics**, *University of Rochester*, Rochester, NY, USA, GPA: 3.9/4.0.
Courses: Syntax I & II, Morphology, Phonetics and Phonology, Field Methods I & II, Historical Linguistics, Data Science for Linguistics

2019–2021 **M.A. English Language and Literature**, *Madras Christian College, University of Madras*, India, First Class with Distinction, CGPA: 7.9.
Courses: English Language Linguistics, English Language Teaching, Translation Studies, Critical Theory

2016–2019 **B.A. English Language and Literature**, *Madras Christian College, University of Madras*, India, First Class, CGPA: 7.3.
Relevant Courses: Grammar and Linguistics, German I & II, Prosody and Rhetoric

Research Experience

University of Rochester, Rochester, NY

2024–present **Research Technician**, Department of Brain and Cognitive Sciences

Advisors Prof. Chigusa Kurumada and Prof. Xin Xie

Research Adaptive speech perception and accented speech production

Corpus ***ADAPT: A Database of Accented Speech Production with Transcriptions***

- Developed and released ADAPT, a database of phonemically annotated native and nonnative English speech, comprising 180 word- and 81 sentence-level productions from 30 talkers (15 American English [AE], 15 Mandarin-accented English [ME]); 8.6 hours of speech across three repetitions per stimulus. Publicly available on OSF; Manuscript under review at *JASA*.
- Evaluated forced-alignment accuracy against manually corrected Penn Forced Aligner output in Praat by quantifying boundary displacement, tolerance-window agreement, and signed and absolute duration differences for fricatives. Only 48.6% of automatic boundaries fell within 20 ms of manual corrections, increasing to 74.5% within 50 ms.
- Trained and supervised 9 undergraduate research assistants in Praat-based phonetic annotation and R-based data visualization; established annotation and quality-control procedures (inter-annotator correlation for segment durations: $r = 0.91$).

Case study ***Distributional Analysis of Fricative Voicing in Mandarin-Accented English***

- Investigated whether reduced phonetic category distinctness in L2 speech reflects increased within-category variability or systematic shifts in category structure, analyzing word-final /s/–/z/ and word-initial /f/–/v/ across 1,243 tokens from 30 talkers.
- Quantified category means, within-category variability, and between-category separability in a rate-normalized, talker-centered vowel–frication duration space; fitted linear models and Gamma generalized linear mixed-effects models (GLMMs) with by-talker random intercepts.
- Found that ME /s/–/z/ categories were more compressed ($p = .001$), without significantly greater within-category variability ($p = .45$), indicating a systematic reduction in contrastive separation rather than increased production noise.
- Found that ME talkers maintained AE-like /f/–/v/ separability through a greater frication-duration contrast, whereas AE talkers relied on both vowel and frication duration cues, revealing differences in cue weighting.
- Presented "Toward Distributional Analyses of Accented Speech Production: The ADAPT Corpus and a Fricative Voicing Case Study" (with S. Cummings, C. Kurumada, and X. Xie) at CAMP, November 2026.

Experimental Design and Data Collection

Targeted Fricative Voicing Database

- Designed fricative word lists targeting voicing contrasts across real words, nonwords, sentences, and Mandarin fricatives stimuli; recorded 20 AE and ME speakers.
- Developed a semi-automated Python workflow for Charsiu forced alignment and Praat-based phonetic annotation.

Mandarin Tone Learning

- Contributed to the design and piloting of a Gorilla-based tone-learning experiment for AE listeners; selected visual stimuli, refined experimental procedures, and developed initial R visualizations of learning and retention.

2022–2024 **Graduate Research Assistant**, Department of Linguistics

Advisor Prof. Maya R. Abtahian

Project **Variable Deletion of the Vowel *Schwa* in Jakarta Indonesian**

- Investigated socio-phonetic and phonological predictors of variable *schwa* deletion in spoken Jakarta Indonesian, examining the roles of adjacent sounds, utterance duration, syllable structure, and etymology.
- Conducted socio-phonetic analysis of 2,000+ audio tokens from a multi-speaker Jakarta Indonesian corpus using ELAN, Praat, and Excel.
- Used spectral-slice analysis to distinguish true schwa deletion from apparent deletion caused by gestural overlap with adjacent segments.

2023–2024 **Graduate Technical Assistant**, Department of Linguistics

Advisor Prof. Joyce McDonough

Project **Archiving Dene Language Data**

- Compiled and organized an audio-video catalog spanning five Dene languages: Dene Suline, Dogrib, North Slavey, South Slavey, and Tsuut'ina.
- Implemented DARLA for semi-automated speech alignment, generating segmented TextGrids to support transcription and formant analysis.
- Prepared archival materials for ELDP, including dataset metadata, IPA transcriptions, and annotated speech data.

Boston University, Boston, MA

Summer 2023 **Research Assistant**, Phonetics, Acquisition, and Multilingualism Lab, Department of Linguistics

Advisor Prof. Charles B. Chang

Project **AlcoLang: Intoxication in Bilingual (German-English) Speech**

- Contributed to AlcoLang, a cross-linguistic project on how alcohol intoxication affects bilingual speech production in L1 (German, Korean, Mandarin) and L2 (English)
- Manually corrected forced-aligned word and phone tiers for 200+ German utterances from dialogue-reading sessions, and transcribed them in X-SAMPA
- Segmented stop consonants and wrote Praat scripts to extract stop durations for analyses of intoxication-related consonant changes

Research Projects

University of Rochester

2023–2024 **Scrambling in Bengali: An A- vs. A'-Movement Distinction**

Advisor Prof. Asia Pietraszko

Published Work
- Developed a syntactic account of Bengali scrambling across four environments, using weak crossover, reciprocal binding, and reconstruction to distinguish A- from A-movement.
- Established that vP-internal scrambling is exclusively A-movement, clause-internal and infinitival scrambling permit both movement types, and cross-finite scrambling is exclusively A-movement; linked these patterns to clause structure and identified speaker variation consistent with A-scrambling across finite clauses.

Current Work
- Found pilot evidence from six native speakers that Bengali objects can A-scramble out of finite clauses, using minimal-pair judgments across weak crossover, reciprocal binding, and Condition C, with controls for wh-movement, island effects, and embedded clause structure.
- Identified a subject–object asymmetry in long-distance extraction: embedded objects can be extracted, but embedded subjects cannot, even when NPI and genitive subjects control for topic parses; found no evidence of hyperraising.

2023–2024 ***Master's Dissertation*: Vowel Hiatus Resolution in Setswana**

Advisors Prof. Nadine Grimm and Prof. Joyce McDonough

- Investigated vowel hiatus resolution at morpheme, syllable, and word boundaries in Setswana
- Identified and described context-dependent strategies such as vowel elision, glide formation, diphthongization, re-syllabification, and glottal stop epenthesis, explaining how they vary across different morphological contexts
- Compiled and annotated a Setswana VHR corpus with 100+ minutes of audio, including detailed segmentations and IPA annotations on PRAAT, Leipzig-glosses and translations, and tonal analysis

Madras Christian College

2020–2021 ***Master's Dissertation*: Dance as a Pragmatic System of Communication with Reference to Rabindranath Tagore's *Tasher Desh***

Advisor Prof. Franklin S. Daniel

- Analysed dance as a multimodal communicative system by applying principles of semantics and pragmatics to understand how meaning is conveyed non-verbally
- Investigated the pragmatic modulation of meaning by examining contextualized semantic cues, including gestures, facial expression, and body language and their role in enhancing interpretive comprehension
- Applied the above analysis methodologies to Rabindranath Tagore's dance-drama *Tasher Desh* (*The Land of Cards*) and demonstrated that a pragmatic analysis broadens the communicative potential of dance

Publications

2026 **Basu, Sreyoshi**, Leslie Famularo, Shawn Cummings, Chigusa Kurumada, and Xin Xie. "Toward Distributional Analyses of Accented Speech Production: The ADAPT Corpus and a Fricative Voicing Case Study." *The Journal of the Acoustical Society of America*. Under review.

2025 **Basu, Sreyoshi.** "Scrambling in Bengali: An A-/A′-Movement Distinction." In *Proceedings of FASAL-14*, 2025.

2025 **Basu, Sreyoshi.** "Phonological Processes at Affix Boundaries in Setswana." *Working Papers in the Language Sciences at the University of Rochester*, vol. 11. Submitted, 2026.

2024 **Basu, Sreyoshi.** "Scrambling in Bengali: An A- vs. A′-Movement Distinction." *Working Papers in the Language Sciences at the University of Rochester*, vol. 10, 2024.

Presentations

2024 Scrambling in Bengali: An A- vs. A'-Movement Distinction. *The 14th (Formal) Approaches to South Asian Languages*. **Sreyoshi Basu**. *Stony Brook University, Stony Brook, NY. April 2024*

2024 Vowel Hiatus Resolution in Setswana. *Phonetics and Phonology in Upstate New York*. **Sreyoshi Basu**. *Cornell University, Ithaca, NY. March 2024*

2019 Understanding Pluralism in Indian Identity as Depicted in Girish Karnad's *Hayavadana*. *The Annual Intercollegiate Students Seminar on Indian Theatre*. **Sreyoshi Basu** and Tirzah David. *Stella Maris College, Chennai, India. September 2019*

Work Experience

June 2023–Aug 2023 **Academic Editorial Intern**, Bloomsbury Publishing, New York, NY

- Managed editorial and administrative tasks across Literary Studies, Film & Media Studies, and Sound & Music Studies lists; edited cover copy and manuscripts for style consistency, accuracy, and quality.
- Coordinated 50+ submissions to Cataloging-in-Publication and Library of Congress programs, maintained internal databases, and liaised with authors and editors to solicit endorsements and manuscript reviews.

Jan 2022–July 2022 **Project Associate**, Indian Institute of Technology Madras, Chennai, India

- Produced multimedia content, metadata, researched articles, and video documentaries on IIT Madras's history and socio-cultural events; supported digital archiving, website documentation, and conferences on archival storytelling.

June 2020–Aug 2020 **Technical Writer**, Riafy Technologies Pvt. Limited, Kerala, India

- Developed the training knowledge base for Feddy, Federal Bank's AI/ML chatbot, desigining 3,500+ variants of customer queries with corresponding solutions using conversational design principles as part of a 7-member team; conducted accuracy and quality assurance testing to improve ML training and user interfaces, and supported API documentation

Service and Mentoring

University of Rochester

2024–present **Mentor**, *Department of Brain and Cognitive Sciences*.
Trained and supervised 9 undergraduate research assistants in Praat-based phonetic annotation and R-based data visualization, across the University of Rochester and the University of California, Irvine

2024–2025 **Editor**, ***Working Papers in the Language Sciences***.
Read submissions, liaised with reviewers, provided anonymous reviewer feedback to authors, and published Vol. 10 of WPLS:UR

Madras Christian College

2020–2021 **Editor**, ***Cornucopia: The Canvas of Plenty***.
Published the 2020–2021 edition of the official annual college magazine. ISBN 978-81-949704-15

Volunteering

Oct 2020–Apr 2021 **Teaching Assistant**, Teach For India, Chennai, India

- Delivered 1-on-1 English literacy support to Grade 4 students during COVID-19, using storytelling, phonics, and interactive word games to address learning gaps and improve reading fluency by four levels, tracked through regular assessments

Certifications

2024 **Data Analysis with R Programming**. Google. August 2024, Coursera

2024 **The Bilingual Brain**. University of Houston. May 2024, Coursera

2022 **Big data and Language 1**. KAIST. January 2022, Coursera

2021 **First Step Korean.** Yonsei University. November 2021, Coursera

Skills and Interests

Languages: English (Native), Hindi (Native), Bengali (Native), Tamil (Fluent), German (Beginner), Korean (Beginner), Setswana (Research experience)

Software: Praat, ELAN, FLEx, R/RStudio, Python, LaTeX

Other Tools: Microsoft Office, Google Workspace, Canva

Professional Skills: Experimental Design, Acoustic Analysis, Data Analysis, Research Documentation, Academic Writing, Copyediting & Proofreading

Interests: Translated Fiction, Indian Classical Dance, Baking, Painting